

## Chapter 6, Problem 2PP

### Problem

If a  $10\text{-}\mu\text{F}$  capacitor is connected to a voltage source with

$$v(t) = 20 \cos(200t) \text{ V}$$

determine the current through the capacitor.

### Step-by-step solution

#### Step 1 of 2

The source voltage connected to the capacitor is,

$$v(t) = 20 \cos(200t) \text{ V}$$

The expression for the current through the capacitor is,

$$i(t) = C \frac{dv(t)}{dt} \dots\dots (1)$$

#### Step 2 of 2

Substitute  $20 \cos(200t) \text{ V}$  for  $v(t)$  and  $10\text{-}\mu\text{F}$  for  $C$  in equation (1).

$$\begin{aligned} i(t) &= 10 \times 10^{-6} \frac{d}{dt} [20 \cos(200t)] \\ &= 10 \times 10^{-6} \times 20 \times 200 [-\sin(200t)] \\ &= -40 \times 10^{-3} \sin(200t) \text{ A} \\ &= -40 \sin(200t) \text{ mA} \end{aligned}$$

Thus, the current through capacitor is  $\boxed{-40 \sin(200t) \text{ mA}}$ .